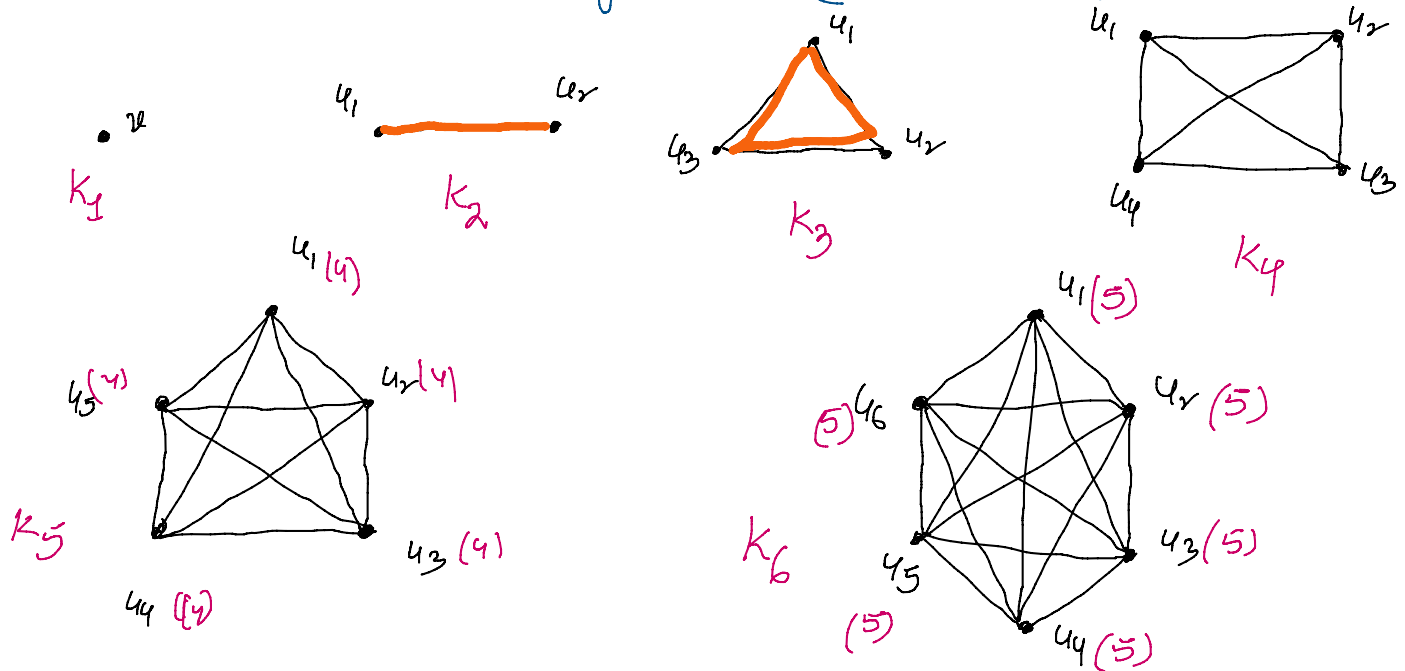


3) Complete Graph $\Rightarrow$ A graph G is s.t.'s Complete graph if every pair of vertices in G are adjacent to each other.

It is denoted by K_n (a complete graph with n vertices)



In K_n

- ① the no. of vertices = n
- ② the degree of every vertex = $n-1$.
- ③ the total no. of edges = $\frac{n(n-1)}{2}$

In K_n ,

the degree of every vertex = $n-1$

By Handshaking

$$\sum \deg(v) = 2e$$

$$(n-1) + (n-1) + (n-1) + \dots + (n-1) = 2e$$

$$n(n-1) = 2e \Rightarrow$$

$$e = \frac{n(n-1)}{2}$$

The no. of edges in $K_5 = \frac{5(5-1)}{2} = 10$

The no. of edges in $K_{10} = \frac{10(10-1)}{2} = 45$

For what value of n , K_n contains an Euler circuit?

- (a) All (b) Even (c) ~~odd~~ (d) No value

In K_n $\deg(v) = n-1 = \text{even}$

$n = \text{odd}$

For what value of n , K_n contains an Hamiltonian path

- (a) Even (b) ~~all~~ (c) odd (d) None.

For what value of n , K_n contains an Hamiltonian circuit?

- (a) ~~All except $n=2$~~ (b) (c) (d)

(4) Regular graph $\rightarrow$ A graph is s.t.b regular if degree of every vertex is same.

If $\deg(v) = k$, $v \in V$ then we call it as k -regular graph.

v

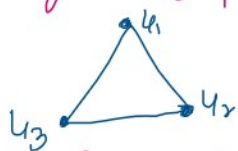
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$v_1 \text{ --- } v_2$

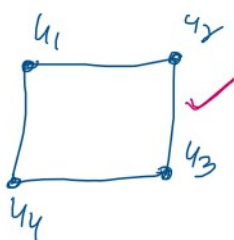
0-regular graph

v

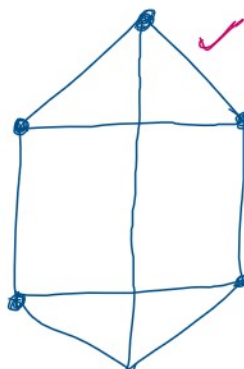
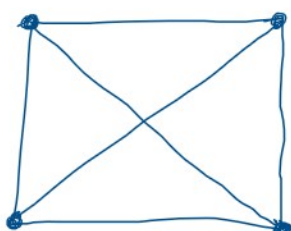
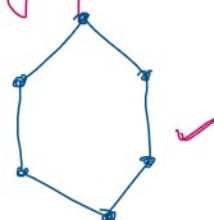
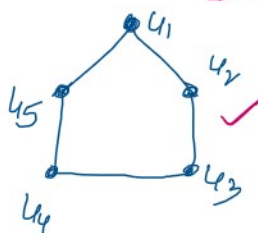


2-regular graph

v_1 v_2



1-regular graph



3-regular graph

All complete graphs are regular

but

all regular graphs are not complete

Ques a 3-regular graph with 3 vertices?

$$\text{Sum of degrees} = 3(3) = 9 \text{ (odd no)}$$

Not possible

How many vertices does a regular graph of degree four with 10 edges have.

$n = ?$

$e = 10$

$$\deg(v) = 4$$

By Handshaking Thm

$$\sum \deg(v) = 2e$$

$$n(4) = 2(10)$$

$$4n = 20$$
$$\boxed{n = 5}$$

The degree sequence of K_8 is

7, 7, 7, 7, 7, 7, 7, 7